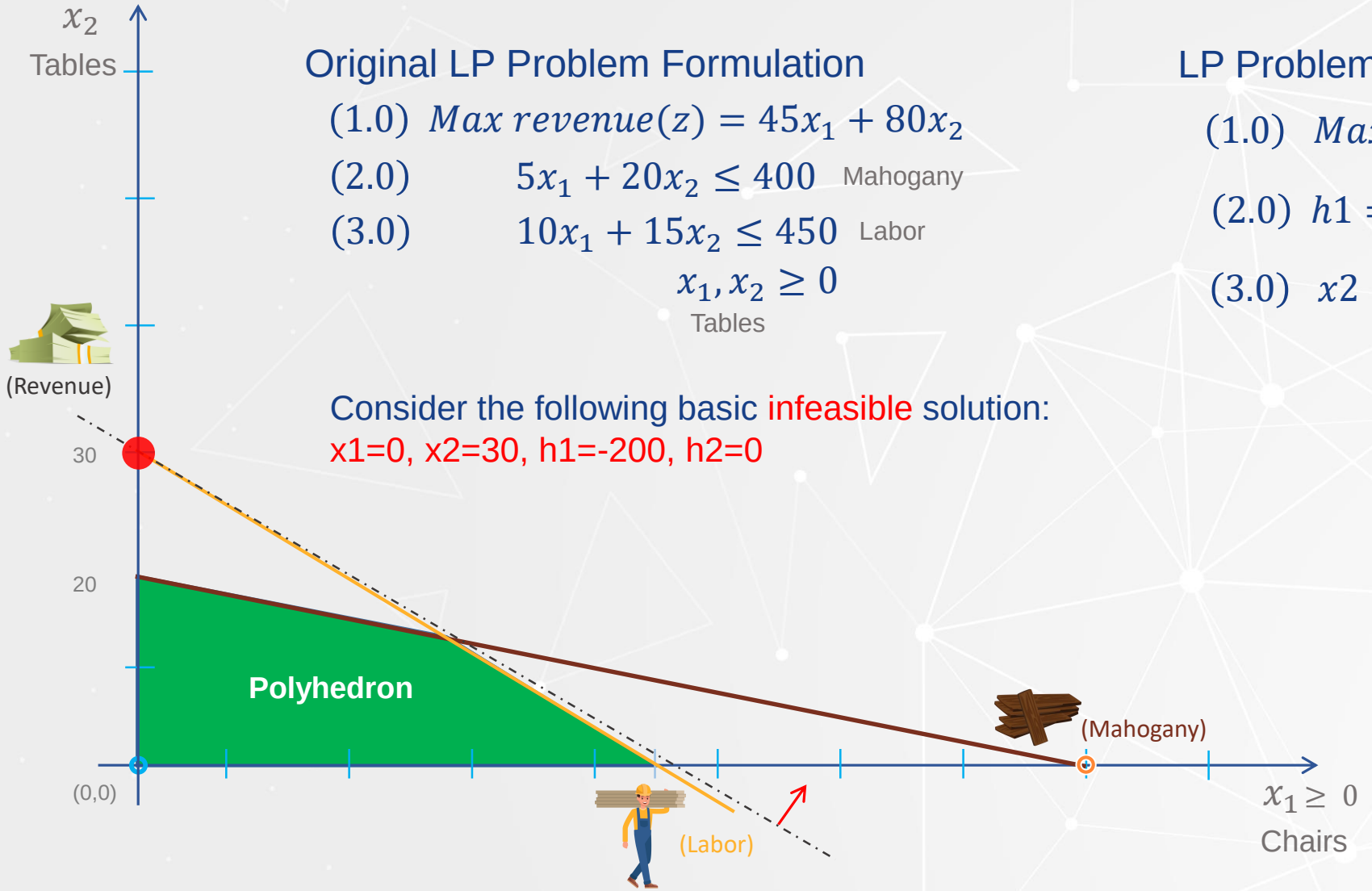
# Dual Simplex method

Variation of  Simplex method

# The Dual simplex method

The key idea of the dual simplex method is to apply the simplex method to the dual problem, but using the canonical form of the primal problem.




Original LP Problem Formulation

(1.0) $Max\ revenue(z) = 45x_1 + 80x_2$

(2.0) $5x_1 + 20x_2 \leq 400$ Mahogany

(3.0) $10x_1 + 15x_2 \leq 450$ Labor

$x_1, x_2 \geq 0$

Tables

LP Problem Formulation in Canonical Form

(1.0) $Max\ z = 2400 - {}^{25}/_{3}\, x1 - {}^{80}/_{15}\, h2$

(2.0) $h1 = -200 + {}^{25}/_{3}\, x1 + {}^{4}/_{3}\, h2$

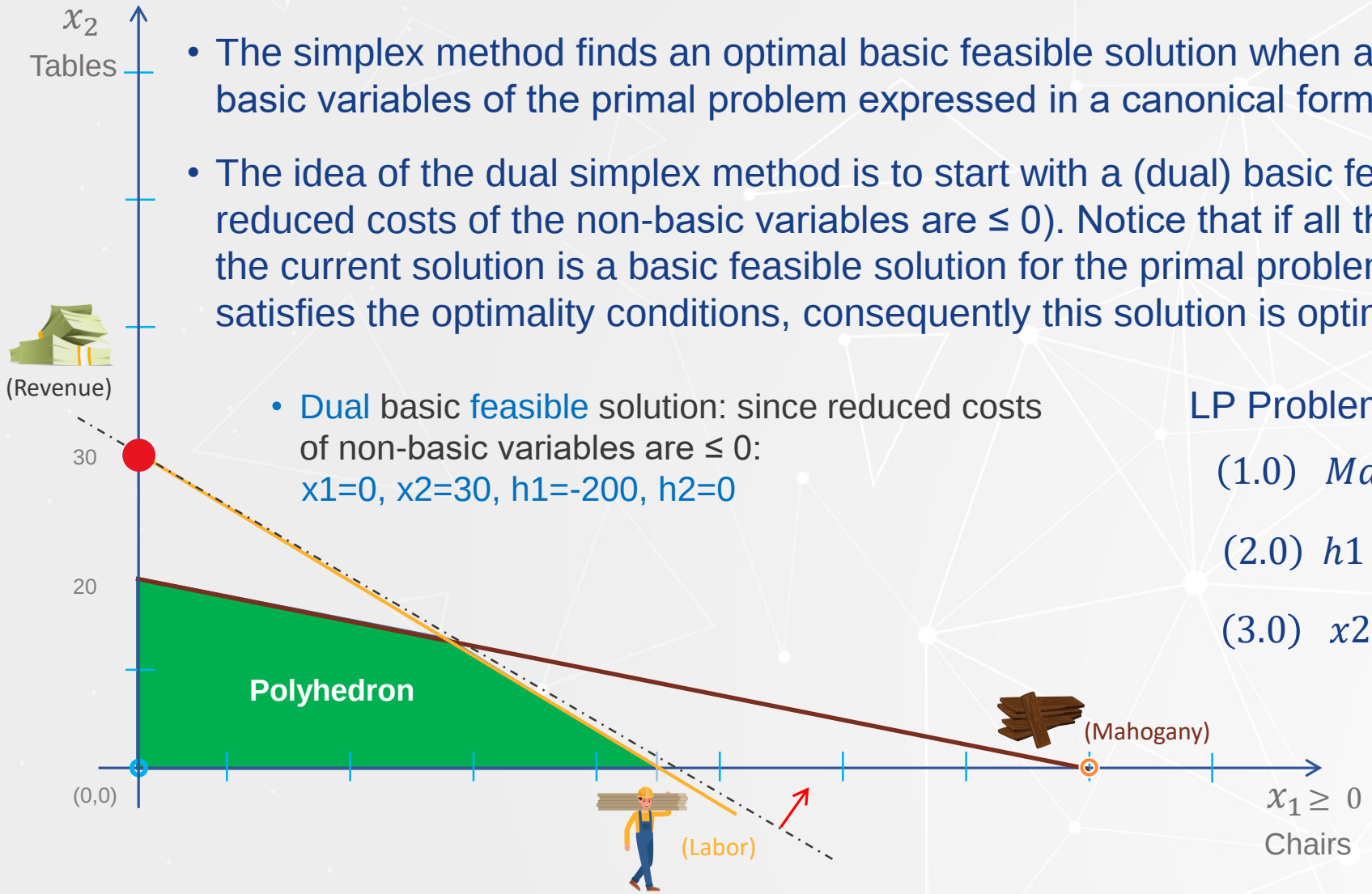
(3.0) $x2 = 30 - {}^{2}/_{3}\, x1 - {}^{1}/_{15}\, h2$

$x1, x2, h1, h2 \geq 0$

Consider the following basic infeasible solution:
x1=0, x2=30, h1=-200, h2=0

# The Dual simplex method ..2

- Recall that the simplex method iterates from one basic feasible solution to another, always trying to improve the value of the objective function.
- The simplex method finds an optimal basic feasible solution when all the reduced costs of the non-basic variables of the primal problem expressed in a canonical form are ≤ 0. (Maximization problem).
- The idea of the dual simplex method is to start with a (dual) basic feasible solution (i.e. all the reduced costs of the non-basic variables are ≤ 0). Notice that if all the basic variables are ≥ 0, then the current solution is a basic feasible solution for the primal problem in a canonical form that satisfies the optimality conditions, consequently this solution is optimal.




- Dual basic feasible solution: since reduced costs of non-basic variables are ≤ 0:
  x1=0, x2=30, h1=-200, h2=0

LP Problem Formulation in Canonical Form

$$(1.0)\quad Max\ z = 2400 - {}^{25}/_{3}\, x1 - {}^{80}/_{15}\, h2$$

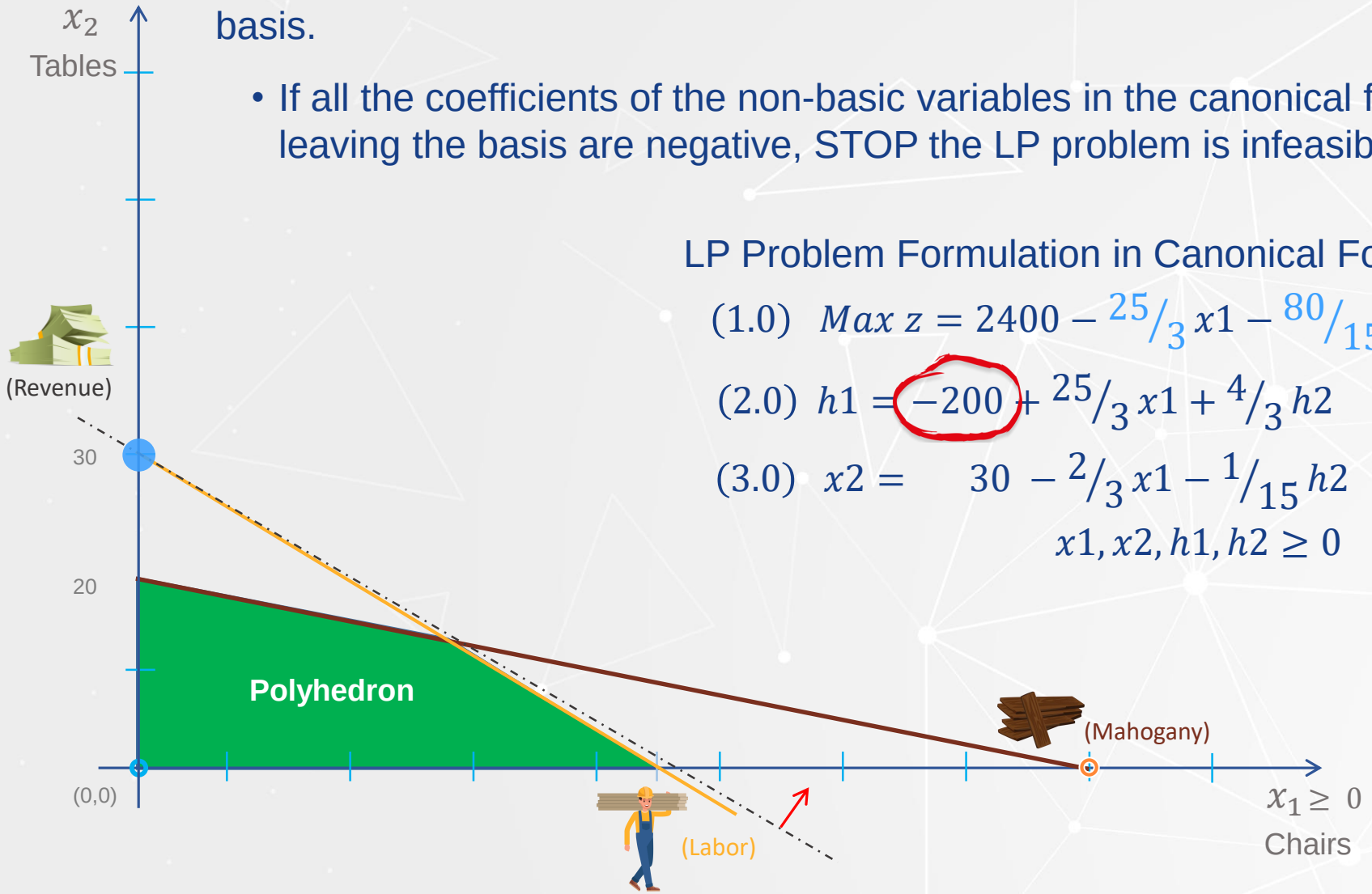
$$(2.0)\quad h1 = -200 + {}^{25}/_{3}\, x1 + {}^{4}/_{3}\, h2$$

$$(3.0)\quad x2 = \quad 30 - {}^{2}/_{3}\, x1 - {}^{1}/_{15}\, h2$$

$$x1, x2, h1, h2 \geq 0$$

# The Dual simplex method ..3

- Suppose that at least there is one basic variable that is negative. Choose the basic variable with the most negative value as the variable to leave the basis. In this example, the variable h1 will leave the basis.
  - If all the coefficients of the non-basic variables in the canonical form equation of the basic variable leaving the basis are negative, STOP the LP problem is infeasible.




LP Problem Formulation in Canonical Form

$$(1.0)\quad Max\ z = 2400 - {}^{25}/_{3}\, x1 - {}^{80}/_{15}\, h2$$

$$(2.0)\quad h1 = -200 + {}^{25}/_{3}\, x1 + {}^{4}/_{3}\, h2$$

$$(3.0)\quad x2 = 30 - {}^{2}/_{3}\, x1 - {}^{1}/_{15}\, h2$$

$$x1, x2, h1, h2 \geq 0$$

# The Dual simplex method ..4

- Else, there is at least one non-basic variable with a positive coefficient, consider the minimum ratio test of the absolute of the reduce costs and the positive coefficient of non-basic variables in the canonical form equation of the basic variable leaving the basis. The non-basic variable with the minimum ratio enters the basis. Determine the new (dual) basic feasible solution. Minimum ratio test:
  - Min { (25/3)/(25/3)=1, (80/15)/(4/3)=4,}= 25/3)/(25/3)=1. Hence, the variable x1 enters the basis.
  - Determine the new (dual) basic feasible solution by pivoting.

LP Problem Formulation in Canonical Form

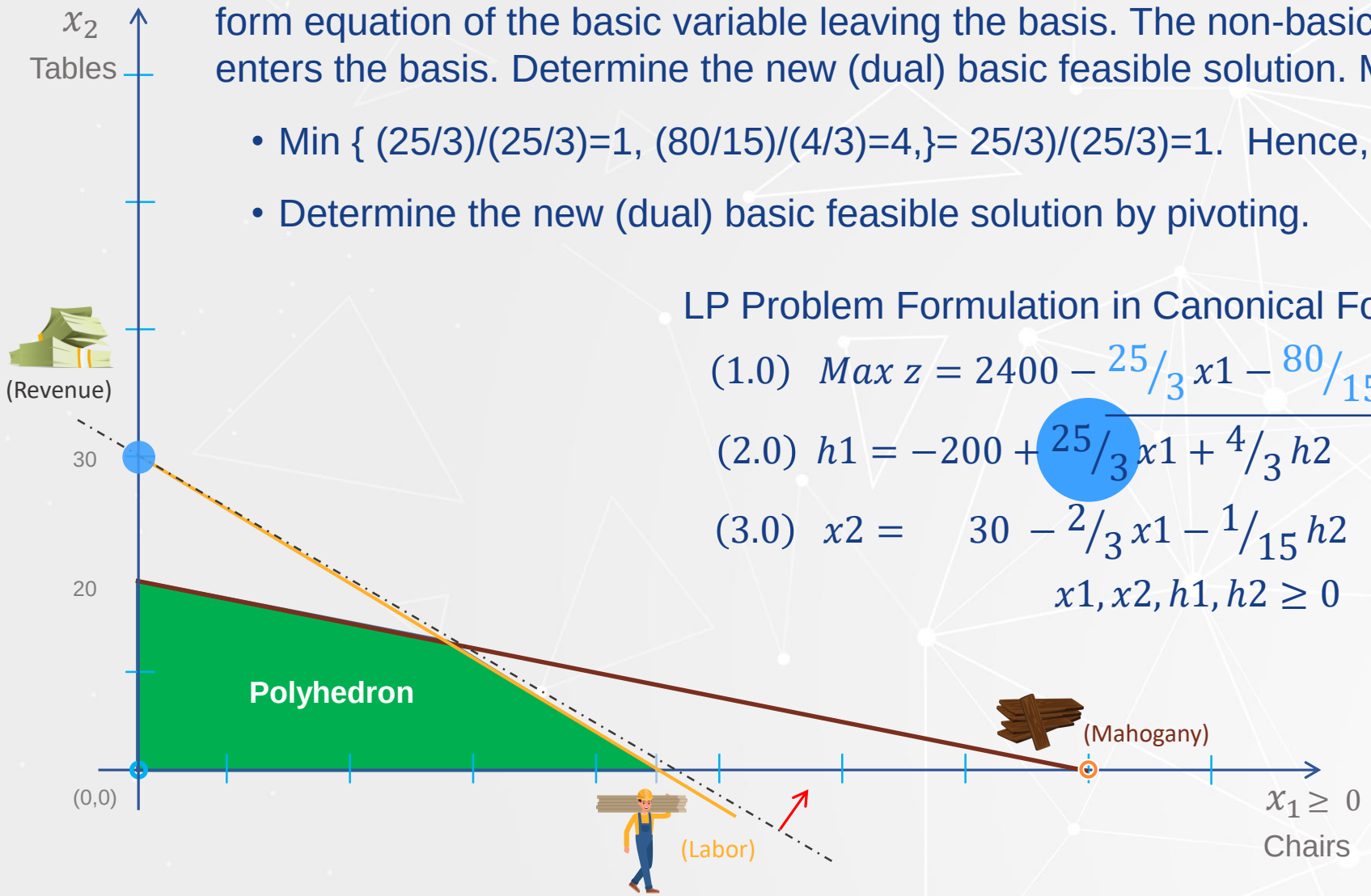
$$(1.0)\quad Max\ z = 2400 - {}^{25}/_{3}\,x1 - {}^{80}/_{15}\,h2$$

$$(2.0)\quad h1 = -200 + {}^{25}/_{3}\,x1 + {}^{4}/_{3}\,h2$$

$$(3.0)\quad x2 = \quad 30 - {}^{2}/_{3}\,x1 - {}^{1}/_{15}\,h2$$

$$x1, x2, h1, h2 \geq 0$$

Pivot



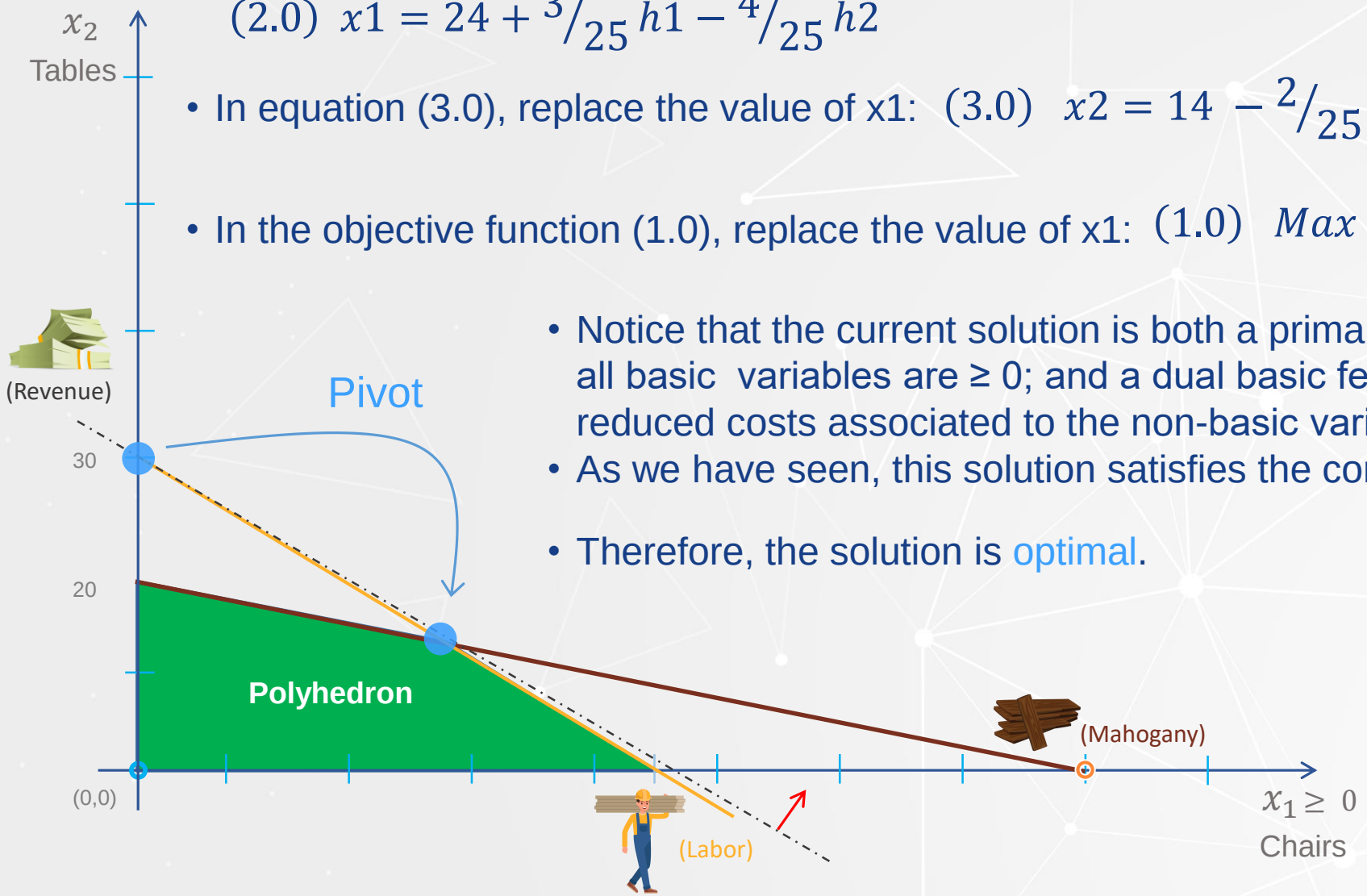
# The Dual simplex method ..5

- Pivoting). In equation (2.0), express basic variable x1 in terms of non-basic variables h1 and h2:

$$(2.0)\ \ x1 = 24 + {}^{3}/_{25}\, h1 - {}^{4}/_{25}\, h2$$

- In equation (3.0), replace the value of x1: $(3.0)\ \ x2 = 14 - {}^{2}/_{25}\, h1 - {}^{1}/_{25}\, h2 \qquad x1, x2, h1, h2 \geq 0$

- In the objective function (1.0), replace the value of x1: $(1.0)\ \ Max\ z = 2200 - 1h1 - 4h2$




- Notice that the current solution is both a primal basic feasible solution, since all basic variables are ≥ 0; and a dual basic feasible solution, since all the reduced costs associated to the non-basic variables are ≤ 0.
- As we have seen, this solution satisfies the complementary conditions:
- Therefore, the solution is optimal.

# The Dual simplex method ..5

- The dual simplex method is recommended for LP problems in which a dual basic feasible solution is available.
- The dual simplex method is particularly useful for reoptimizing an LP problem after a constraint has been added, since we don’t need to start the solution approach from scratch.
- The cutting planes method to solve MIP problems relies heavily on the dual simplex. The core idea of the cutting planes method is to add a constraint to the LP problem continuous relaxation whenever an integer variable has a fractional value in the optimal solution. The added constraint will make this fractional value of the integer variable infeasible.